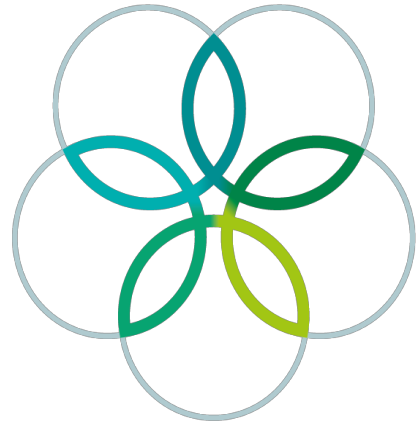
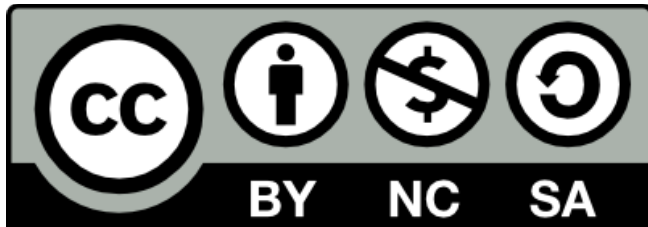


INTERNATIONAL
BIOLOGY
OLYMPIAD e. V.

IBO



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**IBO 1997, Turkmenistan
THEORETICAL TEST
PART A**

1. In plants with flowers (angiosperms), the female gametophyte is very small and consists of
 - A) 20 to 25 nuclei.
 - B) 18 to 20 nuclei.
 - C) 7 to 9 nuclei.
 - D) 1 to 2 nuclei.

2. In which of the following the gametophyte plant is dominant ?
 - A) Rhynophyta
 - B) Bryophyta
 - C) Lycopodiophyta
 - D) Pinophyta
 - E) Equisetophyta

3. A dark -grown oat seedling illuminated from one side with a halogen bulb curves towards the bulb. A colored filter was inserted between the bulb and the seedling and the effect of illumination remained unaffected. What was the color of the filter?
 - A) Red
 - B) Green
 - C) Yellow
 - D) Blue
 - E) Such an effect is not possible. Only full spectrum white light may have the effect described above

4. Dichlorophenol indophenol (DCPIP) is a blue dye that is decolorized when reduced. After being mixed with DCPIP which of the followings would show the greatest change in color?
 - A) isolated chloroplasts in the dark.
 - B) isolated chloroplasts in the light.
 - C) chlorophyll extract in the dark.
 - D) boiled chloroplasts in the dark.
 - E) boiled chloroplasts in the light.

5. Phytochrome far red (PFR) inhibits flowering in:
 - A) Day-neutral plants
 - B) Vernalized plants
 - C) Long-day plants
 - D) Short-day plants
 - E) Senescing plants

6. In a monocot stem, such as corn, there are no distinct areas of cortex and pith. The tissue that performs the functions as the cortex and pith is referred to as;
 - A) Endodermis
 - B) Ground tissue
 - C) Meristem
 - D) Vascular cambium
 - E) Periderm

7. Which of the following techniques would be most suitable for the study of DNA synthesis in a cell?
 - A) Phase contrast microscopy
 - B) Electron microscopy
 - C) Autoradiography
 - D) Serial- sections microscopy
 - E) Nuclear transplantation

8. In poikilotherm animals, oxidation and phosphorylation is physiologic response to the decreasing surrounding temperature. In this condition which of the following structures change its function manner?

- A) Nucleus inner membrane
- B) Mitochondria inner membrane
- C) Mitochondria outer membrane
- D) Hyaloplasma
- E) Peroxisom

9. Which of the followings occurs in the same direction as evolution?

- A) Genetic Drift
- B) Gene Flow
- C) Natural Selection
- D) Mutation
- E) Both Genetic Drift and Gene flow

10. In some localities in Africa there are three different distasteful species of butterfly. Different females of the edible swallow tail butterfly, *Papilio dardanus*, mimic each of the distasteful models. So the initial single population has been separated into three different populations that differ in their color pattern as each has mimicked a different distasteful model. Which of the following explains this event?

- A) Stabilizing selection
- B) Directional selection
- C) Disruptive selection
- D) Bottleneck effect
- E) Founders effect

11. Which of the followings is not accepted as prezygotic isolating mechanism in speciation?

- A) Gametic Isolation
- B) Mechanical Isolation
- C) Hybrid Breakdown Isolation
- D) Ecological Isolation
- E) Temporal Isolation

12. Why are mycoplasmas resistant to penicillin although they are sensitive to tetracycline? (Tetracycline is a kind of antibiotic that inhibits protein synthesis)

- A) Since they lack a cell wall
- B) Since they lack nuclear envelope
- C) Since they are smaller than viruses
- D) Since they are anaerobic
- E) Since their cell walls are made of compounds like protein

13. Which of the following livings causes the black bread mold?

- A) Ascomycota
- B) Basidiomycota
- C) Deuteromycota
- D) Zygomycota

14. Which of the followings are the fungal partners of nonwoody plants in mycorrhiza symbiotic relationships?

- A) Zygomycota
- B) Basidiomycota
- C) Deuteromycota
- D) Ascomycota
- E) Both C and D.

15. Which of the followings controls the amount and kinds of minerals that enter the xylem in the apoplast route of transportation of water?

- A) Semipermeable cell membrane
- B) Plasmodesmata

- C) Endodermis
- D) Cortex parenchyma cells.
- E) Pericycle

16. Which of the followings is the origin of multicellular branch roots?

- A) Epidermis
- B) Cortex
- C) Endodermis
- D) Tracheid
- E) Pericycle

17. In order for a bacterium to produce an eucaryotic protein, which of the followings must first be isolated from an eucaryotic cell prior to cloning?

- A) The protein from the rough endoplasmic reticulum
- B) The protein primary DNA transcript from the nucleus
- C) The protein of mRNA from the cytoplasm
- D) The segments of DNA that control transcription for this protein
- E) The introns from the segment of DNA that codes for the protein

18. In humans the majority of ingested calcium is excreted in the:

- A) Urine
- B) Sweat
- C) Plasma
- D) Feces

19. The old blind man have not noticed the wasp in his cup of juice. It has string in his throat. What is the best help for him?

- A) To warm his neck
- B) To give him some alcohol
- C) To give him some cold water
- D) To give him some sedative
- E) To give him vomitory

20. Which of the following cells recognizes an antigen presented in combination with MHC II? (Major Histocompatibility Complex II)

- A) Suppressor cell
- B) Cytotoxic cell
- C) Neutrophil cell
- D) Helper cell
- E) Basophil cell

21. When humans movement become incorrect disorderly trembling and lose equilibrium and there are pronunciation disorders and speech difficulties the lesion is found in:

- A) Spinal cord
- B) Thalamus
- C) Hypothalamus
- D) Medulla oblongata
- E) Cerebellum

22. Which of the following structures is formed due to the transformation of the lateral organ?

- A) Pressure receptors
- B) Semicircular canals
- C) Middle ear
- D) Cochlea
- E) Statocyst

23. Which of the followings depicts the pattern flow of energy and materials through an ecosystem?

- A) Food chain
- B) Biomass pyramid
- C) Numbers pyramid

- D) Food web
- E) A and D

24. Into a large, tightly stoppered flask filled with nitrogen 100ml of live yeast made suspension in minimal medium containing 0.1 mole/l glucose as carbon source was introduced. After 24 hours somewhat more than 100 ml of carbon dioxide (as measured under standard conditions) accumulated in the flask (assume no carbon dioxide dissolves in the medium). What was the approximate concentration of glucose in the medium at that moment?

- A) Almost zero
- B) 0.01 mole/l
- C) 0.05 mole/l
- D) 0.075mole/l
- E) 0.09 mole/l

25. The principal constitute of myelin is

- A) Carbohydrate
- B) Lipid
- C) Mucopolysaccharide
- D) Nucleic acid
- E) Protein

26. Various substances can affect the excitability of neuron. Calcium ions make a neuron less excitable by:

- A) Increasing the amount of sodium ions that enter the voltage gated sodium channels
- B) Binding to the neurolemma, making it less permeable to potassium ions thus limiting repolarization and reducing the action potential.
- C) Binding to the proteins that make up the sodium channels thus reducing the amount of sodium ions that can enter.
- D) Binding to the active site of the sodium potassium pump which causes repolarization at a voltage lower than -70 mV
- E) Reacting with the negatively charged chlorine ion to form calcium chloride which neutralizes the membrane potential.

27. In certain terrestrial animals the body surface has evolved as a respiratory structure. This is characterized by

- A) The presence of deep indentations in the body surface to facilitate the diffusion of gas deep into the body.
- B) The exchange of gas directly between all the body cells and the atmosphere
- C) Special enzymes that allow for active transport of oxygen into the body and carbon dioxide out of the body.
- D) The counter current exchange of carbon dioxide and oxygen between the air and mucus covering the animals' body surface.
- E) The presence of blood vessels close to the surface to receive and transport oxygen to other body regions.

28. The function of RUGAE in the stomach is to _____ .

- A) Crush the food
- B) Increase the capacity of the stomach.
- C) Secrete amylase
- D) By muscular contractions, mix the food with gastric juices
- E) Increase the surface area for secretion

29. Which statement is false in comparing Metanephridia with protonephridia?

- A) Metanephridia have a tuft of cilia to move fluids along while protonephridia only have a single flagellum.
- B) The metanephridium is open at both ends
- C) Metanephridia are surrounded by capillaries which permits useful materials to be reabsorbed and returned to the circulation
- D) Useful materials are reabsorbed in the metanephridia.
- E) Water is conserved and urine is concentrated

30. In a colony of honey bees, the dominance of the queen is maintained by the

- A)** Provision of eggs by the queen
- B)** Secretion of pheromones by the queen
- C)** Suppression of sexual development of the workers
- D)** None of the above

31. Substitution of a nitrogen base in a gene can result in:

- A)** Nonsense mutation
- B)** Frame shift mutation
- C)** Missense mutation
- D)** Silent mutation
- E)** All kinds of the above mentioned mutations except B

32. The frequency of crossovers between linked genes A and B is 30% between B and C is 40%, between C and D is 45%, between A and C is 10%, between B and D is 5%. The sequence of genes on chromosome is :

- A)** ABCD or DCBA
- B)** BDCA or ACDB
- C)** CABD or DBAC
- D)** DBCA or ACBD
- E)** DCAB or BACD

33. Some mammals have no collar bone in their shoulder girdle. A typical feature of these mammals is;

- A)** Being arboreal
- B)** Being aquatic
- C)** Being capable of soaring or active flight
- D)** Having mobility of forelimbs limited to one plane only
- E)** Neither of the features given above

34. How old are the earliest known fossils of the genus Homo ?

- A)** Several hundred thousand years
- B)** 1-9 million years
- C)** 10-19 million years
- D)** 20-100 million years
- E)** More than 100 million years

35. Which of the following makes up the inner bark of a woody plant?

- A)** Primary phloem
- B)** Secondary phloem
- C)** Primary xylem
- D)** Secondary xylem
- E)** Pith

36. Which of the following statements about the upper and lower epidermis is correct?

- A)** Lower epidermis cells are living parenchyma cells while upper epidermis cells are not.
- B)** Upper epidermis cells' wall on the outside of the leaf is thicker than on the inside while it is not valid for lower epidermis.
- C)** Upper epidermis cells are relatively transparent while lower epidermis cells contain chloroplasts.
- D)** The upper epidermis has a thicker cuticle than the lower epidermis.
- E)** The lower epidermis mostly contains lots of trichomes while there is no trichome in the upper epidermis.

37. The resemblance between one orchid species and female wasps is so strong that male wasps mount the flowers and attempt to copulate with them. This is an example of:

- A)** Parasitism
- B)** Coevolution
- C)** Artificial selection
- D)** Adaptive radiation
- E)** Selective breeding

38. Fruits that form from a single flower that contains many separate carpels, are referred to as:

- A)** Accessory fruits
- B)** Aggregate fruits
- C)** Dry fruits
- D)** Multiple fruits
- E)** Simple fruits

39. How do shade-avoiding plants respond when they absorb more far-red light?

- A)** They produce more leaves
- B)** They grow additional parts
- C)** They branch extremely
- D)** They grow taller
- E)** They stop growing

40. Plants respond to wounding by herbivorous insects by producing a plant polypeptide known as _____ that may trigger the plant to produce proteinase inhibitors that disrupt insect digestion.

- A)** Florigen
- B)** Polyamine
- C)** Phytoalexin
- D)** Salicylic acid
- E)** Systemin

41. Which of the following models maintains that populations diverge from one another by the slow accumulation of adaptive characteristics within each population?

- A)** Punctuated equilibrium
- B)** Paedomorphosis
- C)** Hybridization
- D)** Gradualism
- E)** Allopolyploidy

42. Proteins synthesized in E-coli have which of the followings at their terminal end?

- A)** N- formyl- methionine
- B)** N- acetyl- adenine
- C)** Adenine triphosphate
- D)** The AUG codon
- E)** The UUU codon.

43. Which of the followings are the ancestors of mitochondria and chloroplasts according to endosymbiosis theory respectively?

- A)** Purple sulfur bacteria, cyanobacteria
- B)** Purple non-sulfur bacteria, cyanobacteria
- C)** Purple sulfur bacteria, green algae
- D)** Purple non- sulfur bacteria, green algae
- E)** None of the above

44. Catabolite activator protein _____ .

- A)** Increases the affinity of the promoter region for RNA polymerase
- B)** Decreases RNA polymerase activity
- C)** Becomes nonfunctional when it binds to cAMP
- D)** Both A and B are true
- E)** Both A and C are true

45. The lactose operon regulates the synthesis of _____ .

- A)** b - galactosidase
- B)** Galactoside transacetylase
- C)** Galactose transferase
- D)** Both A and B
- E)** A, B, C are correct

46. TATA box is seen in _____ cells and is the site where _____ .

- A) Prokaryotic; RNA polymerase binds
- B) Eucaryotic cells; DNA ligase cleaves introns
- C) Multicellular eucaryotes, RNA polymerase binds
- D) All of the above are true
- E) None of the above is true

47. How may recombinant genes be introduced into monocotyledons?

- A) By the use of Ti vectors
- B) By the use of plasmids as vectors
- C) By the use of genetic shotguns
- D) By the use of radioactive probes
- E) None of the above

48. Which of the followings can be used as DNA vectors?

- A) Bacteriophages
- B) Plasmids
- C) E- coli cells
- D) Both A and B
- E) Both A and C

49. Immunoglobulins, designated _____ increase antimicrobial activity in mucus.

- A) Ig A
- B) Ig E
- C) Ig G
- D) Ig M
- E) Ig Z

50. Antibody- mediated responses work best against _____ .

- A) Intracellular pathogens.
- B) Extracellular pathogens
- C) Toxins
- D) B and C
- E) All of the above.

51. Which of the followings is wrong?

- A) The peroxysomes in plant cells are also called glyoxysomes
- B) Peroxysomes and glyoxysomes contain enzymes that degrade fatty acids
- C) Peroxysomes are present in the photosynthetic cell of green plants
- D) Microbodies are cytochemically defined by their catalase activity

52. Which of the followings is true?

- A) Dynein is a Ca^{++} ATPase and generates the energy for the bending of the ciliurn and flagellum.
- B) The initial polymerization of tubulin often occurs at certain regions of the cell termed microtubule organizing centers (MTOCs).
- C) If cells are microinjected with Ca^{++} microtubule disassembly is blocked and assembly is promoted.
- D) Desmin is a kind of microfilament.
- E) Desmin is not found in cardiac or smooth muscle.

53. Which of the following statements is wrong?

- A) The nuclear lamina is composed of four different polypeptides termed lamins.
- B) The lamins are intermediate filaments.
- C) Nuclear envelope contains structural proteins but does not contain any enzymes.
- D) Small molecules enter the nucleus more readily than do large ones.
- E) All of them is true

54. In eucaryotes there are commonly three different RNA polymerases, which act at different promoters to transcribe the various kinds of RNA in the cell. Which of the followings about the RNA polymerase I is true in eucaryotes?

- A) It occurs in the nucleoplasm and synthesizes mRNA
- B) It occurs in the nucleolus and synthesizes mRNA
- C) It occurs in the nucleolus and synthesizes tRNA
- D) It occurs in the nucleoplasm and synthesizes mRNA
- E) It occurs in the nucleolus and synthesizes rRNA

55. All chordates have _____ in some stage of their life.

- A) A spinal chord
- B) Gills
- C) A notochord
- D) Vertebrae
- E) Hair

56. A cladogram is based on:

- A) Homologous characteristics
- B) Analogous characteristics
- C) Polyphyletic taxa
- D) Fossil evidence only
- E) Derived characters only

57. The phenetic system is based primarily on:

- A) Known evolutionary relationships
- B) Fossil evidence
- C) Shared characteristics
- D) Ancestral characteristics
- E) Both ancestral and shared characteristics

58. A typical R- selected organism might be a _____ : K- selected organism might be a

- A) Mouse, elephant
- B) Elephant, mouse
- C) Mouse, rabbit
- D) Rabbit, fly
- E) Rabbit, mouse

59. _____ is the least frequent of the dispersal patterns seen in nature and may be seen in situations in which the environment is particularly homogeneous.

- A) Aggregated distribution
- B) Clumped distribution
- C) Uniform distribution
- D) Density - dependent distribution
- E) Random dispersion

60. When some animals mature they then migrate long distances to their breeding grounds or to escape harsh weather. A signal for this type of behavior would be:

- A) A conditioned stimulus
- B) The length of day
- C) Negative reinforcement
- D) A sign stimulus

61. For an animal to respond to a pheromone it must:

- A) Have a specific receptor
- B) Be able to see it coming
- C) Be able to smell it
- D) Have a thin epidermis or hair so it can penetrate
- E) Be of the opposite sex of the individual from which it was produced

62. In polymerase chain reaction technology the two strands of DNA are separated by:

- A) Increasing temperature
- B) Treating them with restriction enzymes

- C) Centrifugation
- D) Increasing pH
- E) Both A and C are true

63. Which of the following areas of the world produce the most O₂ per year?

- A) Temperate forests
- B) Grasslands
- C) Lakes and streams
- D) Oceans

64. The concentration of phytoplankton is significantly lower in open seas than in coastal waters. It is commonly believed that this is caused directly by:

- A) Insufficient abundance of mineral nutrients
- B) Too low concentration of dissolved carbon dioxide
- C) Insufficient mixing of water
- D) Massive presence of phytoplanktophagous fish
- E) Too high salinity

65. Lianas and epiphytes are especially common in:

- A) Tropical rain forests
- B) Savannah
- C) Temperate grassland
- D) Taiga

66. Lichens are most typically formed by the symbiotic association of an alga or cyanobacterium and a(n) :

- A) Ascomycota
- B) Deuteromycota
- C) Basidiomycota
- D) Zygomycota
- E) Spirochete

67. The logistic equation:

- A) Models an S-shaped growth curve
- B) Takes into account the carrying capacity
- C) Shows continued exponential growth
- D) Both A and B
- E) Both B and C

68. At which of the given times should the CO₂ concentration in the forest be at the highest level each day during the summer?

- A) Just before sunrise
- B) Noon
- C) Just before sunset
- D) Midnight

69. Clumped dispersion is typically influenced by :

- A) Social behavior such as family groups
- B) Asexual reproduction in plants or invertebrates
- C) Limited seed dispersal in plants
- D) A patchy distribution of resources
- E) All of these

70. The area of greatest seasonal ozone depletion is in the:

- A) Stratosphere above the Arctic
- B) Stratosphere above the Antarctic
- C) Troposphere above the Arctic
- D) Troposphere above the Antarctic
- E) Lithosphere above the Arctic

71. Which of the followings about the golgi complex is false?

- A) The golgi complex under the electron microscope is composed of flattened sacs termed cisternae
- B) Sulfation of proteins doesn't occur in the golgi
- C) Glycosylation of proteins occur in the golgi
- D) The trans region of golgi faces the plasma membrane and has larger vesicles than cis region of golgi

72. Studying cells using a phase contrast microscope gives the advantage because:

- A) The resolving power is greater
- B) The magnification is greater
- C) It uses a beam of electrons which can penetrate the tissue of living organisms, thus allowing one to see the organelles enclosed by the plasma membrane
- D) Organelles of cells can be seen without staining which usually results in the death of cells

73. The scanning electron microscope differs from the transmission electron microscope because the scanning electron microscope:

- A) Can view a living specimen
- B) Relies on the emission of secondary electrons emitted from a gold-coated object
- C) Can view the internal structure of a cell
- D) Utilizes a beam of light that can pass through the specimen
- E) Gives a two dimensional image of the object being studied

74. Typically, which grows the most slowly?

- A) Bread moulds
- B) Lichens
- C) Yeast
- D) Mushrooms
- E) Sac fungi (Ascomycota)

75. Which of the followings is characterized by cold temperatures little rain and few or no trees?

- A) Taiga
- B) Savannah
- C) Tundra
- D) Desert

76. Which of the following biomes has the greatest species diversity?

- A) Tropical Forest
- B) Savannah
- C) Grassland
- D) Temperate deciduous forest

77. The tropical rain forest is characterized by:

- A) High density of plants, but nutrient poor soil
- B) High density of plants, and nutrient rich soil
- C) Low density of plants due to nutrient poor soil
- D) Low density of plants, but nutrient rich soil
- E) None of the above responses are correct

78. Which the following pairs is mismatched?

- A) Behavior; response to environment signals
- B) Capacity for behaviour; is learned
- C) Behaviour; can be modified
- D) Persistent changes in behaviour; learning
- E) Behaviour; adaptation for survival

79. In a place where number of individuals establish adjacent territories if one individual gets further from the center of its territory and approaches the edge of the next, it:

- A) Encounters a big fight on entering the next territory
- B) Loses control of its mates

- C) Passes a neutral line between territories where neither is dominant
- D) Becomes more aggressive as it gets closer to the adjacent territory

80. Which of the followings about the plasma membranes is not true?

- A) In most plasma membranes the two membrane leaflets (cytoplasmic and exoplasmic leaflets) have same lipid compositions
- B) Membrane proteins bind asymmetrically to the two layers of lipid
- C) Most membrane proteins and lipids are laterally mobile in the membrane
- D) Cell coat (glycocalyx) is found at the exoplasmic face

81. Which of the followings about the cell junctions is false?

- A) Many plant cells are interconnected by plasmodesmata which allow metabolites with low molecular weight and signaling molecules to move between cells
- B) Ca^{++} ion concentration affects the closure of gap junctions
- C) Tight junctions consist of connexin molecules; six formed in a hexagon cylinder in one plasma membrane joined to six arranged in the same array in the adjacent cell membrane
- D) Spot desmosomes are found in all epithelial cells

82. Which one of the followings is false for mitochondria?

- A) The size and coding capacity of mitochondria DNA varies in different organisms
- B) Mitochondria contain their own genetic material
- C) Mitochondria ribosomes are different from the cytoplasmic ribosomes
- D) Most mitochondrial proteins are synthesized in the mitochondria
- E) Plant mitochondria DNA's are larger than the other organisms' mitochondria DNA

83. Which of the following plant hormones delay senescence or aging?

- A) Auxin
- B) Cytokinin
- C) Ethylene
- D) Gibberellin
- E) Abscissic acid

84. Which of the following phylum's members cause sleeping sickness?

- A) Rhizopoda
- B) Zoomastigina
- C) Ciliophora
- D) Foraminifera
- E) Actinopoda

85. The beginning of cephalization is first seen in the:

- A) Scyphozoans
- B) Poriferans
- C) Platyhelminthes
- D) Nematodes
- E) Nemertean

86. In the body, although nerve impulses can travel in both directions along an axon impulses are usually unidirectional. This is because:

- A) Dendrites have a higher threshold than axons
- B) The axon terminals have a higher threshold than axons
- C) The myelin sheath directs the flow of ions toward the axon terminals
- D) The axon terminals contain neurotransmitter receptors and the dendrites do not
- E) Dendrites contain no neurotransmitter substance

87. Which of the followings about the archaeobacteria is false?

- A) They all lack peptidoglycan in their cell walls
- B) Their lipids contain an ether linkage between the glycerol and the fatty acid residues
- C) Translation is initiated by N-formyl methionine t RNA
- D) They live in swamps

88. In chemiosmosis hydrogen ions (protons) release their energy to produce ATP as they pass through _____.

- A) The outer mitochondrial membrane
- B) ATP synthase
- C) ATP dehydrogenase
- D) ATP decarboxylase
- E) A group of electron carriers

89. Tetanus toxin inhibits the breakdown of acetylcholine at the myoneural junction. This effects muscle contraction by _____.

- A) Not allowing the muscle to contract
- B) Preventing the muscle from relaxing
- C) Preventing calcium from entering the muscle
- D) Inhibiting potassium from leaving the muscle
- E) Stopping the formation of actomyosin

90. Protein leaves the interstitial fluid and enters the circulatory system:

- A) Through pores on the arterial side of a capillary bed
- B) Through pores on the venous side of a capillary bed
- C) By being phagocytized and transported to the blood through eosinophils
- D) Via the lymphatics which deposits them into the subclavian vein
- E) By active transport mediated by permeases in the capillary epithelium

91. In the citric acid cycle, two acetyl CoA molecules are metabolized to:

- A) $2 \text{ CO}_2 + 2 \text{ ATP} + 2 \text{ NADH} + 2 \text{ FADH}$
- B) $4 \text{ CO}_2 + 6 \text{ NADH} + 2 \text{ FADH}_2 + 2 \text{ ATP}$
- C) Fructose 1,6 biphosphate
- D) $\text{Glucose} + 2 \text{ CO}_2 + 2 \text{ NADH} + 2 \text{ FADH}_2 + 2 \text{ ATP}$
- E) 2 PGAL

92. Which of the followings is true?

- A) Lactose operon in E. coli is only controlled by negative control by the lactose repressor
- B) When E.coli is grown in the presence of both glucose and lactose, both glucose and lactose is utilized
- C) Catabalite gene activator protein (CAP) has no influence on transcription until ATP has bound to it
- D) Different proteins that are coded by a same mRNA molecule are always produced in similar numbers
- E) DNA repair genes at many different sites on the chromosome are regulated by only a single repressor in prokaryots

93. A mRNA "cap" :

- A) Marks the mRNA for degradation
- B) Serves as the attachment point for a poly-A tail
- C) Prevents translation
- D) Decreases the half - life of the mRNA
- E) None of the answers is correct

94. Which of the followings can be found on a completed bacterial mRNA?

- A) Intron region
- B) A leader sequence
- C) A primer sequence
- D) Poly A tail
- E) None of them

95. In an E-coli bacterium, there are three types of DNA polymerase which are called DNA polymerase I, DNA polymerase III and DNA polymerase II. Which of the followings about the functions of these enzymes is wrong?

- A) All of them have 5' ® 3' polymerization activity
- B) Only DNA polymerase II has not the 5' ® 3' exonuclease activity

- C) Only DNA polymerase I and III have the 3' → 5' exonuclease activity
- D) DNA polymerase I has the capacity to remove the RNA primers
- E) DNA polymerase III makes up the Okazaki fragments

96. Which of the followings is not correct for CAM plants?

- A) Crassulacean Acid Metabolism is found in fewer than 5 percent of plants
- B) Although the cell could not metabolize normally if the acid accumulated in the cytoplasm, throughout the night more and more malic acid is produced and stored in the cell's vacuole, building a reservoir of fixed CO₂
- C) Malic acid is broken down into CO₂ and pyruvate during day-time. It is then that CO₂ becomes fixed to RBP
- D) CAM synthesis helps to conserve precious water by separating the time of CO₂ fixation; the light reactions, and the Calvin cycle

97. Which of the following amino acids enter the Krebs cycle via being converted to ketoglutaric acids?

- A) 2- carbon amino acids
- B) 3- carbon amino acids
- C) 4- carbon amino acids
- D) 5- carbon amino acids

98. Intolerance to cold and myxedema are typically seen in _____ .

- A) Hyperthyroidism
- B) Hypothyroidism
- C) Acromegaly
- D) Cretinism

99. What's the effect of insulin in adipose tissue?

- A) It stimulates glucose uptake
- B) It stimulates lipolysis
- C) It stimulates hormone-sensitive lipase and lipoprotein lipase
- D) It stimulates urea synthesis

100. The largest white blood cell in humans which is a part of the body's macrophage defense system is:

- A) Neutrophil
- B) Lymphocyte
- C) Monocyte
- D) Eosinophil
- E) Basophil